

Flexible loss functions for binary classification in gradient-boosted decision trees: An application to credit scoring

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ABSTRACT

This paper introduces new flexible loss functions for binary classification in Gradient-Boosted Decision Trees (GBDT) that combine Dice-based and cross-entropy-based losses and offer link functions from either a generalized extreme value (GEV) or exponentiated exponential logistic (EEL) distribution. Testing 27 different GBDT models using XGBoost on a Freddie Mac mortgage loan database showed that the choice of the loss function is useful. Specifically, when the class imbalance ratio (IR) is less than 99, using a skewed GEV distribution-based link function in XGBoost enhances discriminatory power and classification accuracy while retaining a simple model structure, which is particularly important in credit scoring applications. In cases where class imbalances are severe, typically between IRs of 99 and 200, we found that an advanced loss function, which is composed of a symmetric hybrid loss function and a link derived from a positively skewed EEL distribution, outperforms other XGBoost variants. Based on our findings, the accuracy improvements of these proposed extensions result in lower misclassification costs, which are especially evident when IR is below 99, which results in higher profitability for the business. Furthermore, the study highlights the transparency associated with GBDT, which is also an integral component of financial applications. Researchers and practitioners can use these insights to create more accurate and discriminative machine learning models, with possible extensions to other GBDT implementations and machine learning techniques that take into account loss functions. The source code for the proposed approach is publicly available at <https://github.com/jm-ml/flexible-losses-for-binary-classification-with-GBDT>.

1. Introduction

Gradient boosting algorithms are popular in machine learning due to their ability to achieve high predictive power and scalability to diverse problems. In essence, gradient boosting involves constructing an ensemble predictor by iteratively combining weak models (base learners) greedily (Friedman, 2001). Among the popular base learners are decision trees due to their simplicity and non-parametric nature. Gradient boosting with decision trees is known as Gradient-Boosted Decision Trees (GBDT). GBDT has been implemented in several ways, the most recent of which include XGBoost (T. Chen & Guestrin, 2016), LightGBM (Ke et al., 2017) and CatBoost (Dorogush, Ershov, & Gulin, 2018).

Recent machine learning competitions focusing on classification tasks have been won by ensembles that use GBDT. For instance, XGBoost alone won the Recommender Systems 2020 challenge, while a combination of XGBoost and deep neural networks won the 2021 challenge (Deotte, Liu, Schifferer, & Titericz, 2021; Schifferer, Titericz, Deotte,

Henkel, Onodera, Liu, & Erdem, 2020). Additionally, recent Women in Data Science Datathon challenges have been won by ensembles dominated by XGBoost and LightGBM (Cohen, Dagan, Cohen-Inger, Ofer, & Rokach, 2021). In the premier annual Knowledge Discovery in Databases competition held in 2022, LightGBM was among the top three winning solutions for classification tasks (Wu, Liu, Gazo, Bedrich, & Qu, 2022; Zhang, Zheng, Yan, Li, Hu, Zhou, & Liu, 2022). These achievements demonstrate the effectiveness of GBDT for classification tasks.

The use of GBDT in classification tasks is, however, restricted by the use of a classic cross-entropy loss in conjunction with a logit link. Focussing on a binary 0/1 classification problem, the logit link function first converts model predictions into a number between 0 and 1, before the cross-entropy loss quantifies how close the probabilistic predictions are to the class label. There are several limitations associated with binary class cross-entropy. These include bias towards the majority class in class-imbalanced data (Jadon, 2020), inability to distinguish between “easy-to-classify” and “hard-to-classify” cases (Lin, Goyal, Girshick, He, & Dollár, 2017), and symmetric loss components for each class (Li,

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Kamnitsas, & Glocker, 2019). A logit link also has a major limitation, which is that it is symmetric around 0.5. This means that the probabilities approach 0 and 1 at the same rate (M.-H. Chen, Dey, & Shao, 1999). When one class is underrepresented, the logit becomes inappropriate since the probabilities of the minority group will be underestimated (Calabrese, Marra, & Angela Osmetti, 2016).

To improve the predictive power of GBDT in class-imbalanced data settings, two generalizations are proposed in this study. We first adapt the Unified Focal (UF) loss function by Yeung, Sala, Schönlieb, and Rundo (2022), to GBDT. The UF loss contains several loss functions designed for handling class-imbalance data learning as special cases, thus eliminating the need to try multiple loss functions. Secondly, we propose the use of flexible link functions that can be symmetric, positively or negatively skewed, depending on the classification problem. We examine both link functions derived from a generalized extreme value distribution (Coles, Bawa, Trenner, & Dorazio, 2001; Kotz & Nadarajah, 2000), as well as those derived from a new class of generalized logistic distributions (Ghosh & Alzaatreh, 2018). The proposed changes aim to provide GBDT with more flexible loss functions, especially for the handling of class imbalance data.

The rest of the paper is organized as follows: Section 2 discusses related works and places the study in the context of the current literature. Section 3 provides an overview of XGBoost for binary classification, which will be used as a reference implementation of GBDT. In Section 4, we unpack the loss-and-link function extensions that we are proposing. To gain a better understanding of the influence of class imbalance on our proposed extensions, a simulation study is performed in Section 5. We then apply our models to real-life data in Section 6. Finally, Section 7 concludes the study.

2. Related works

In the literature, there are three common approaches for modelling class imbalance in datasets: algorithmic modifications, data rebalancing, or a combination of both (Krawczyk, 2016). One leading approach to algorithmic modification is to penalize the misclassification of minority cases, such as in Xia, Liu, and Liu (2017), Zhou, Yao, Wen, Wang, Zhou, He, and Liang (2017), or (Kim et al., 2019). On the other hand, data rebalancing methods focus on oversampling the minority class to balance the dataset before training a model. Synthetic Minority Over-sampling Technique (SMOTE) is a well-established data rebalancing technique that generates artificial minority cases (Fernández, García, Herrera, & Chawla, 2018). In recent years, Generative Adversarial Neural Networks (GANs) have gained prominence for their ability to rebalance data by creating fake minority cases (Sampath, Maurtua, Aguilar Martin, & Gutierrez, 2021).

Batista, Bazzan, and Monard (2003) further proposed cleaning the rebalanced data using undersampling techniques such as edited nearest neighbours (ENN) and Tomek links (TL). With ENN, cases whose class labels differ from the majority of their K-nearest neighbours (NN) are removed from the data. Comparatively, the TL method uses the 1-NN rule to locate pairs of opposite-class labels that are closest to one another and removes only instances belonging to the majority class. On the other hand, hybrid methods, such as balanced random forest (BRF) (O'Brien & Ishwaran, 2019) and EasyEnsemble (Liu, Wu, & Zhou, 2008), combine algorithmic and data rebalancing approaches by training ensembles of CART (L Breiman, Friedman, Olshen, & Stone, 1984) and AdaBoost (Schapire, 1999), with re-balanced bootstrap samples.

This study aims to implement algorithmic modifications to GDBT, taking the XGBoost implementation as a starting point. XGBoost has also been modified algorithmically in the following studies. Xia et al. (2017) proposed using a cost-sensitive link function in XGBoost for credit scoring. The rationale for using this approach is that in credit scoring, it is more costly to grant credit to a borrower who ends up defaulting than the opportunity cost of rejecting a potentially good borrower. Wang, Deng, and Wang (2020) presented a Python package called Imbalance-

XGBoost, which gives users the option to use either a weighted cross-entropy or a Focal loss in XGBoost. Trisanto, Rismawati, Mulya, and Kurniadi (2021) proposed a modified version of the Focal loss function that incorporates class weights for use in XGBoost. In our previous work (Mushava & Murray, 2022), we reported superior results by pairing a modified Focal loss with a link function based on the standard GEV distribution in XGBoost.

In the context of highly class-imbalanced data, selecting the right loss function is crucial to ensure that even the edge cases are taken into account. One example of this is image segmentation problems, which are often associated with high-class imbalance. To address this issue, specialized loss functions have been developed to train artificial neural networks in computer vision applications (Jadon, 2020). Image segmentation involves assigning a label to every pixel in an image so that pixels with the same label share similar characteristics. For example, in medical image segmentation for oncology, the tumour is typically much smaller than its host organ, resulting in a high-class imbalance. Recently, Yeung et al. (2022) proposed a UF loss to handle class-imbalanced medical image segmentation. The UF loss is a hybrid of generalizations of Dice and cross-entropy loss functions, which eliminates the need to test multiple loss functions independently. The flexible loss function employed in this study is based on the UF loss.

Most researchers focus on customizing loss functions, but little attention has been paid to customizing the associated link function. In binomial regression, link functions are derived from the inverse cumulative distribution function (c.d.f) of a random distribution, such as the logit, which corresponds to the inverse c.d.f of the standard logistic distribution (Prasetyo, Kuswanto, Iriawan, & Ulama, 2020). In this study, flexible link functions for GBDT are proposed that can be negatively or positively skewed, or symmetrical with light or heavy tails. Specifically, we examine links derived from the generalized extreme value (GEV) distribution and a new class of generalized logistic distributions derived from the exponentiated exponential logistic (EEL) distribution. With the GEV distribution, based on extreme value theory, one can fit the parameters that best explain the probability distribution of extreme/rare events (Coles et al., 2001; Kotz & Nadarajah, 2000). On the other hand, the EEL distribution is a new class of generalized logistic distribution proposed by Ghosh and Alzaatreh (2018). It is obtained by mixing two distributions: the exponentiated exponential and the logistic distribution using the technique proposed by Alzaatreh, Lee, and Famoye (2013).

GBDT implementations do not automatically calculate the partial derivatives of the loss function, also known as gradient statistics, that are required for the splitting rules in the decision tree. Consequently, constructing gradient statistics algebraically is a challenging task in adapting the UF loss to GBDT, either with a link function derived from the GEV or EEL distributions. To facilitate reusability and extensibility, we have published the gradient statistics on GitHub, which can be applied to other GDBT implementations such as CatBoost and LightGBM.

The main contributions of this paper can be summarized as follows:

- Adaptation of a hybrid UF loss function to XGBoost, which reduces the requirement for testing multiple loss functions,
- Integration of GEV and EEL distribution-based link functions into the UF loss to allow more flexibility in specifying the symmetry or asymmetry of the links,
- The derivation and publication of algebraic gradient statistics applicable to GBDT in general.

3. Xgboost overview

The following is a brief overview of the XGBoost implementation of GBDT focusing on two class problems. Using strategies such as One-versus-All (OVA) or One-versus-One (OVO), most binary classification problems can be transformed into multi-class classification problems (Murphy, 2012, p. 503).

Let $\mathbf{x}_i = (x_{1i}, x_{2i}, \dots, x_{pi})$ be a p -dimensional vector where x_{ji} denotes the attribute of the feature x_j for a given case i . In our binary classification problem, a variable y_i will be set to 1 if case i was observed to default on their loan (positive class) in a given outcome window, otherwise y_i will be set to 0 (negative class). Consequently, a model development sample with N cases takes the form: $S = \{\mathbf{x}_i, y_i\}_{i=1}^N$. The sub-script i is occasionally dropped whenever the context is clear.

The XGBoost model is structured as follows:

$$\hat{y} = \sum_{t=1}^T f_t(\mathbf{x}) \quad (1)$$

where f_t denotes regression tree base learners that need to be trained sequentially over T rounds/iterations. The objective function that needs to be minimized in each iteration is as follows:

$$L^{(t)} = \sum_{i=1}^N l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \quad (2)$$

where $l(\bullet)$ is a loss function and $\Omega(\bullet)$ is a regularization term. At each iteration t , the goal is to add $f_t(\mathbf{x}_i)$ to the current model $\hat{y}_i^{(t-1)}$ that minimizes the loss compared to the true class y_i for all training samples i . Simultaneously, to avoid over-fitting,

$$\Omega(f_t) = \gamma L + \frac{1}{2} \lambda \sum_{j=1}^L w_j^2 \quad (3)$$

Controls the complexity of the regression trees. This complexity can be measured using L , which represents the number of leaves in the tree, w , which represents the weight assigned to each leaf, as well as γ and λ , which are real-valued parameters.

It can be shown using a second-order Taylor approximation and removing terms independent of f_t that the objective function (2) can be simplified to:

$$L^{(t)} \cong \sum_{i=1}^N g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) + \Omega(f_t) \quad (4)$$

where $g_i = \frac{\partial l(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i}$ and $h_i = \frac{\partial^2 l(y_i, \hat{y}_i^{(t-1)})}{\partial^2 \hat{y}_i}$ are the first and second-order partial derivatives of the loss function, respectively.

Letting I_j denote the number of cases that land in leaf j of f_t , objective (4) can be re-written as a quadratic function of the leaf scores w_j :

$$\tilde{L}^{(t)} = \sum_{j=1}^L \left[\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) w_j^2 \right] + \gamma L \quad (5)$$

It can be shown by differentiating equation (5) with respect to w and setting the result to 0 that the optimal leaf weights are:

$$w_j^* = - \frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} = - \frac{G_j}{H_j + \lambda} \quad (6)$$

where $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$. Therefore, gradient statistics g_i and h_i , also known as the gradient and hessian, respectively, are central to the customisation of the loss functions.

Back-substituting (6) into (5) gives the optimal loss score for a tree structure q as:

$$\tilde{L}^{(t)}(q) = - \frac{1}{2} \sum_{j=1}^L \frac{G_j^2}{H_j + \lambda} + \gamma T \quad (7)$$

where the lowest value is sought. Therefore, the impurity for a single node/leaf can be quantified using:

$$-\frac{1}{2} \frac{G_j^2}{H_j + \lambda} + \gamma$$

The gain (or reduction in the impurity) of parent node I after its split

into a left child node I_L and right child nodes I_R is

$$\begin{aligned} \text{gain} &= \left(-\frac{1}{2} \frac{G_{j \in I}^2}{H_{j \in I} + \lambda} + \gamma \right) - \left[\left(-\frac{1}{2} \frac{G_{j \in I_L}^2}{H_{j \in I_L} + \lambda} + \gamma \right) + \left(-\frac{1}{2} \frac{G_{j \in I_R}^2}{H_{j \in I_R} + \lambda} + \gamma \right) \right] \\ \Rightarrow \text{gain} &= \frac{1}{2} \left[\frac{G_{j \in I_L}^2}{H_{j \in I_L} + \lambda} + \frac{G_{j \in I_R}^2}{H_{j \in I_R} + \lambda} - \frac{G_{j \in I}^2}{H_{j \in I} + \lambda} \right] - \gamma \end{aligned} \quad (8)$$

XGBoost constructs each base learner tree by recursively partitioning the data using splitting rule (8) until a pre-defined stopping rule. To find that variable $x_j \in \{x_1, x_2, \dots, x_p\}$ and its attribute that leads to the largest gain when used to split a node, XGBoost uses either an exact greedy algorithm or an approximate algorithm (see T. Chen and Guestrin (2016) for further details).

4. Proposed flexible loss and link functions

GBDT algorithms such as XGBoost rely on the following cross-entropy loss function for binary classification:

$$l(y_i, \hat{y}_i) = -\frac{1}{N} \sum_{i=1}^N \{y_i \ln \hat{p}_i + (1 - y_i) \ln(1 - \hat{p}_i)\} \quad (9)$$

where $\hat{p}_i$ is the conversion of the model prediction $-\infty < \hat{y}_i < \infty$ to probability via the standard logistic link function:

$$\hat{p}_i = p(y_i = 1 | \hat{y}_i) = \frac{1}{1 + e^{-\hat{y}_i}} \quad (10)$$

It can be shown that the gradient and hessian of the loss (9) with link (10) are $g_i = \hat{p}_i - y_i$ and $h_i = \hat{p}_i(1 - \hat{p}_i)$, respectively. This section presents more flexible loss and link functions as well as the associated gradient statistics.

4.1. Flexible loss functions

Loss functions are mathematical representations of the objective of a training problem. Choosing the correct loss function is vital, especially when dealing with highly imbalanced data, to ensure that all cases, including those at the edges, are considered. Using cross-entropy and Dice-based losses, we will create a hybrid function, which is more adaptable to the classification task at hand.

4.1.1. Cross entropy-based loss function

The general form of the binary class cross-entropy-based loss functions is given by the following modified/weighted Focal loss function (Lin, Goyal, Girshick, He, & Dollár, 2017)

$$l_{mF}(y_i, \hat{y}_i) = -\frac{1}{N} \sum_{i=1}^N \{y_i \alpha_1 (1 - \hat{p}_i)^{\gamma_1} \ln \hat{p}_i + (1 - y_i) \hat{p}_i^{\gamma_1} \ln(1 - \hat{p}_i)\} \quad (11)$$

where $\alpha \in \mathbb{R}$ controls the class weights and $\gamma \in \mathbb{R}$ controls the degree of down-weighting of the loss of the well-classified (i.e. "easy-to-classify") cases. Cases are well-classified if the predicted probability of their assigned class label exceeds 0.5.

The binary cross entropy (9) is the special case: $\alpha_1 = 1$ and $\gamma_1 = 0$. Setting $\gamma_1 = 0$ in (11) produces a weighted binary cross entropy while setting $\alpha_1 = 1$ gives the standard Focal loss function. Fig. 1 illustrates the effect of the parameters α_1 and γ_1 by varying them in turn for the positive class, $y = 1$.

To avoid the suppression of the loss contribution of the minority class, Li, Kamnitsas, and Glocker (2019) suggested removing the focal parameter from the loss function component that relates to the minority class. By doing so, equation (11) becomes asymmetric, thereby prolonging minority cases at the decision boundary, viz

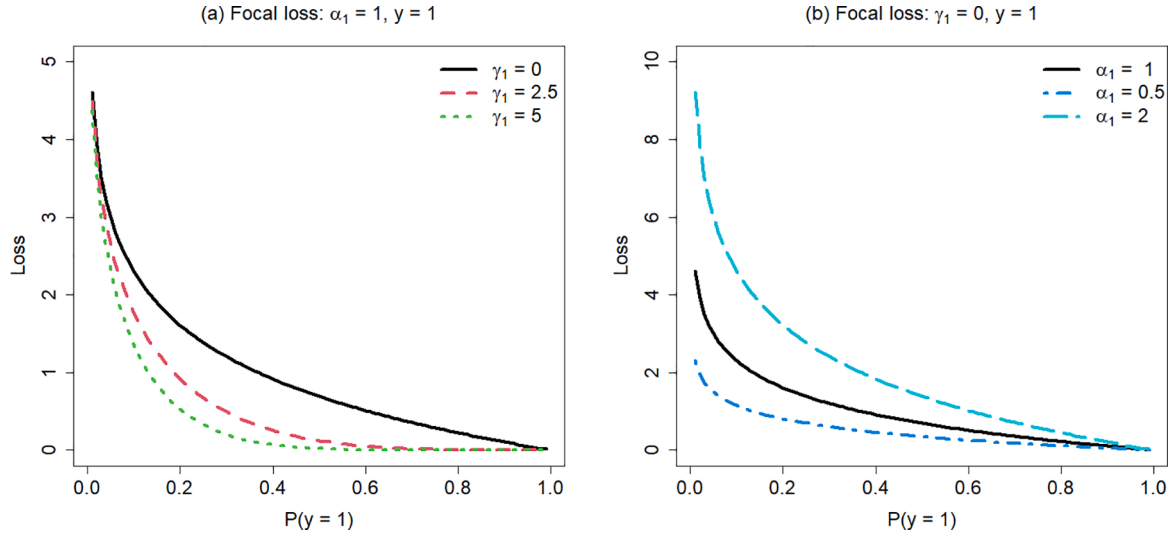


Fig. 1. Relative to the binary cross-entropy (black solid line): a) as $\gamma_1 > 0$ increases, the more the loss function focuses on “hard-to-classify” cases and b) when $\alpha_1 > 1$, the loss function assigns a higher penalty to the misclassification of the positive class and vice versa.

$$l_{maF_0}(y_i, \hat{y}_i) = -\frac{1}{N} \sum_{i=1}^N \{y_i \alpha_1 (1 - \hat{p}_i) \ln \hat{p}_i + (1 - y_i) \hat{p}_i^{\gamma_1} \ln(1 - \hat{p}_i)\} \quad (12)$$

As a further experiment, we will use an asymmetric loss function to remove well-classified minorities from the decision boundary, so that the model focuses on hard-to-classify minority cases, viz

$$l_{maF_1}(y_i, \hat{y}_i) = -\frac{1}{N} \sum_{i=1}^N \{y_i \alpha_1 (1 - \hat{p}_i)^{\gamma_1} \ln \hat{p}_i + (1 - y_i) \hat{p}_i \ln(1 - \hat{p}_i)\} \quad (13)$$

4.1.2. Dice-based loss function

The general form of the Dice-based loss function is given by the following Focal Tversky loss function (Abraham & Khan, 2019)

$$L_{FT} = \sum_{k \in \{0,1\}} (1 - TI_k)^{\frac{1}{\gamma_2}} \quad (14)$$

where $\gamma_2 \in \mathbb{R}$ is a focal parameter and

$$TI_k = \frac{\sum_{i=1}^N y_{ik} \hat{p}_{ik}}{\sum_{i=1}^N y_{ik} \hat{p}_{ik} + \alpha_2 \sum_{i=1}^N y_{ik} (1 - \hat{p}_{ik}) + \beta \sum_{i=1}^N (1 - y_{ik}) \hat{p}_{ik}} \quad (15)$$

is the Tversky index associated with class k (Salehi, Erdogmus, & Gholipour, 2017). In the Tversky index, $y_{ik} \hat{p}_{ik}$ are true positives (TP), $y_{ik} (1 - \hat{p}_{ik})$ are false negatives (FN), and $(1 - y_{ik}) \hat{p}_{ik}$ are false positives (FP). Therefore (15) does not take into account true negatives (TN): $(1 - y_{ik})(1 - \hat{p}_{ik})$. L_{FT} is asymmetric by design $TI = 0$ when $y_{ik} = 0$. Dice-based loss functions are beneficial because they allow for focusing on one particular class during the model training process.

Setting $\gamma_2 = 1$ gives the classical Tversky loss function and adding $\alpha_2 = \beta = 0.5$ simplifies it to the Dice loss function, i.e.

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (16)$$

Fig. 2 illustrates the effect of the parameters γ_2 , α_2 and β on the Focal Tversky loss function by varying them in turn.

4.1.3. Hybrid loss function

The hybrid loss function combines the cross-entropy and Dice-based losses as follows:

$$L_{UF} = \pi L_F + (1 - \pi) L_{FT} \quad (17)$$

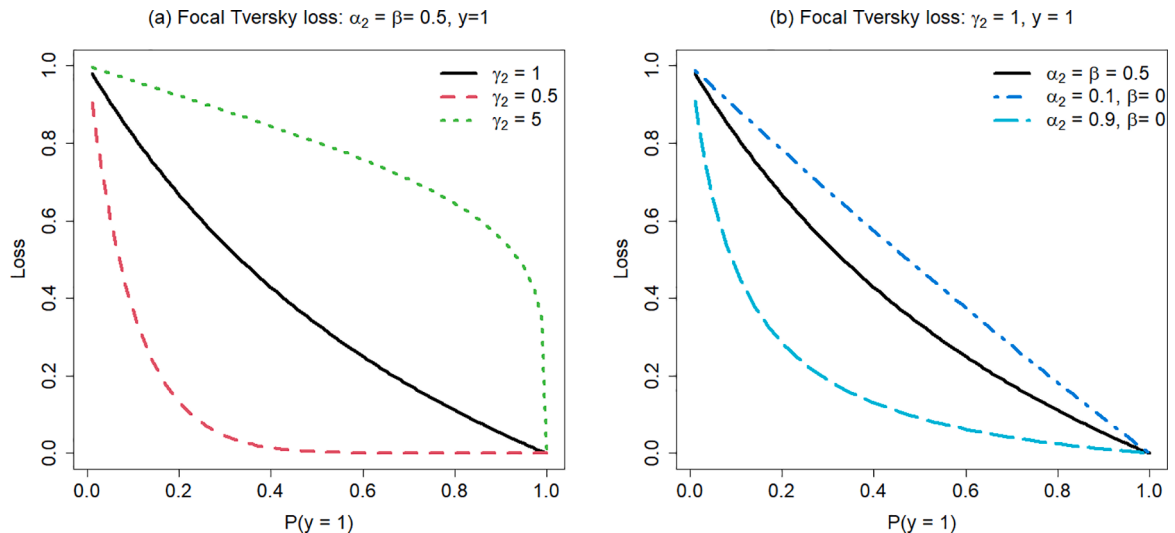


Fig. 2. Relative to the dice loss (black solid line): a) when $0 < \gamma_2 < 1$, the loss function focuses more on hard-to-classify cases, whereas when $\gamma_2 > 1$ it enhances the loss of easy-to-classify cases, and b) setting $\beta > \alpha_2$ assigns more penalty to false negatives and is useful when there is a high level of class imbalance.

where $L_F \in \{l_{mF}, l_{maF_0}, l_{maF_1}\}$ denotes any choice of the Focal loss function versions in (11)-(13), L_{FT} the Focal Tversky loss (14), and $0 \leq \pi \leq 1$ determines the relative weight of the component losses. It follows that the hyperparameter set of the hybrid loss function is $\Phi = \{\alpha_1, \alpha_2, \gamma_1, \gamma_2, \beta, \pi\}$.

Yeung et al. (2022) proposed a so-called UF loss function for semantic segmentation that combines functionally equivalent hyperparameters in a hybrid of the Focal loss and the Focal Tversky. A hyperparameter is a parameter that must be specified before the model can be trained. Firstly, the UF loss function replaces hyperparameters α_1 , α_2 and β with a common hyperparameter δ such that $\alpha_1 = \delta$ in Focal loss function (11), and that the Tversky index (15) be formulated as

$$TI_k^* = \frac{\sum_{i=1}^N y_{ik} \hat{p}_{ik}}{\sum_{i=1}^N y_{ik} \hat{p}_{ik} + \delta \sum_{i=1}^N y_{ik} (1 - \hat{p}_{ik}) + (1 - \delta) \sum_{i=1}^N (1 - y_{ik}) \hat{p}_{ik}} \quad (18)$$

Secondly, Yeung et al. (2022) reformulated (14) to

$$L_{FT} = \sum_{k \in \{0,1\}} (1 - TI_k^*)^{\gamma_2} \quad (19)$$

so that a common focal hyperparameter can be used in both the cross-entropy and Diced-based losses (i.e. $\gamma_1 = \gamma_2 = \gamma$). In this way, the UF loss function uses only three hyperparameters; δ to control the weight of the two classes, γ the focal parameter, and π to determine the weight of the component loss functions.

This study will also utilize the Tversky index (18), but with a loss formulation (14) and allow the other hyperparameter to vary for greater flexibility. Therefore, our UF loss optimizes five hyperparameters: $\Phi^* = \{\alpha_1, \delta, \gamma_1, \gamma_2, \pi\}$. Hereafter, UF is the hybrid loss with (11), while UFO and UF1 are the hybrid losses with (12) and (13), respectively.

4.2. Flexible link functions

In binomial regression, the probability $p = p(y = 1)$ is mapped to the response outcome $-\infty < h(\mathbf{x}) < \infty$ via a link function F^{-1} as follows:

$$F^{-1}(p) = h(\mathbf{x}) \quad (20)$$

where F^{-1} is the inverse cumulative density function (c.d.f) of a random variable $\mathbf{X}$. The link function F^{-1} is also known as the quantile function. For example, the logit link is derived from the c.d.f of the standard logistic distribution, $F(p) = \frac{1}{1+e^{-p}}$. Using the fact that $F(F^{-1}(p)) = p$ and solving for $F^{-1}(p)$:

$$\text{Logit} : F^{-1}(p) = \ln\left(\frac{p}{1-p}\right) \quad (21)$$

In the context of XGBoost, $h(\mathbf{x}) = \hat{y}$ in (1). Therefore (10) comes from the mapping

$$\ln\left(\frac{p}{1-p}\right) = \hat{y}p = \frac{1}{1+e^{-\hat{y}}} \quad (22)$$

The same approach is used in this section to derive links based on the generalized extreme value (GEV) and exponentiated exponential logistic (EEL) distributions.

4.2.1. GEV distribution-based link function

Suppose the predictions $\hat{y}$ in (1) follow a GEV distribution with a parameter set $\{\tau, \mu, \sigma\}$, where $\tau \in \mathbb{R}$ is a shape parameter, $\mu \in \mathbb{R}$ is a location parameter and $\sigma > 0$ is a scale parameter. The probability density function (p.d.f) of $\hat{y}$ is given by:

$$f_{GEV}(\hat{y}) = \frac{1}{\sigma} t(\hat{y})^{\tau+1} e^{-t(\hat{y})} \quad (23)$$

where

$$t(\hat{y}) = \begin{cases} \left(1 + \tau \left(\frac{\hat{y} - \mu}{\sigma}\right)\right)^{-\frac{1}{\tau}}, & \tau \neq 0 \\ e^{-\left(\frac{\hat{y} - \mu}{\sigma}\right)}, & \tau = 0 \end{cases} \quad (24)$$

Nested within the GEV distribution are three types of extreme value distributions. The type I case is the Gumbel family of distributions given by the parameterization $\tau \rightarrow 0$. The type II (Fèchet family) and type III (Weibull family) classes of the extreme value distribution correspond to $\tau > 0$ and $\tau < 0$, respectively.

We propose using the standard GEV distribution ($\mu = 0$ and $\sigma = 1$) with $\tau \neq 0$ to get the c.d.f. of (23) as

$$F_{GEV}(x) = e^{(-[1+\tau x])^{-\frac{1}{\tau}}} \quad (25)$$

Fig. 3 demonstrates the impact of the shape parameter τ on the p.d.f and c.d.f of the standard GEV distribution.

Denoting the probabilistic predictions based on the link derived from a standard GEV distribution as p_{GEV} , the quantile function of the standard GEV distribution is

$$F_{GEV}^{-1}(p) = \frac{(-\ln p_{GEV})^{-\tau} - 1}{\tau} \quad (26)$$

Mapping (26) to $\hat{y}$ and solving for p_{GEV} gives:

$$p_{GEV} = e^{(-[1+\tau \hat{y}])^{-\frac{1}{\tau}}} \quad (27)$$

4.2.2. EEL distribution-based link function

Suppose the predictions $\hat{y}$ in (1) follow an exponentiated-exponential logistic (EEL) distribution with a parameter set $\{\beta, \lambda, \mu, \sigma\}$, where $\beta, \lambda > 0$ are shape parameters, $\mu \in \mathbb{R}$ is a location parameter and $\sigma > 0$ is a scale parameter. The probability density function (p.d.f) of $\hat{y}$ is given by

$$f_{EEL}(\hat{y}) = \frac{\beta \lambda e^{\left(\frac{\hat{y}-\mu}{\sigma}\right)}}{\sigma \left(1 + e^{\left(\frac{\hat{y}-\mu}{\sigma}\right)}\right)^{\lambda+1}} \left(1 - \left(1 + e^{\left(\frac{\hat{y}-\mu}{\sigma}\right)}\right)^{-\lambda}\right)^{\beta-1} \quad (28)$$

The p.d.f. in (28) has the following special cases: $\beta = \lambda = 1$ which leads to the logistic distribution, $\lambda = 1$ which gives the Type I logistic distribution and $\beta = 1$ which yields the Type II logistic distribution.

We propose using the standard EEL distribution ($\mu = 0$ and $\sigma = 1$) to generate the c.d.f

$$F_{EEL}(x) = \left(1 - (1 + e^x)^{-\lambda}\right)^{\beta} \quad (29)$$

Fig. 4 demonstrates the impact of the shape parameters β, λ on the p.d.f and c.d.f of the standard EEL distribution.

Denoting the probabilistic predictions based on the link derived from a standard EEL distribution as p_{EEL} , the quantile function of the standard EEL distribution is

$$F_{EEL}^{-1}(p) = \ln\left(1 - \left(1 + p_{EEL}^{\frac{1}{\beta}}\right)^{-\lambda}\right)^{\frac{1}{\lambda}} \quad (30)$$

Mapping (30) to $\hat{y}$ and solving for p_{EEL} gives:

$$p_{EEL} = \left(1 - (1 + e^{\hat{y}})^{-\lambda}\right)^{\beta} \quad (31)$$

4.3. Derivation of gradient statistics

The derivation of gradient statistics is a non-trivial step in implementing the extensions proposed in this study. This paper is accompanied by a supplementary document providing an algebraic derivation of gradient statistics. Furthermore, we have coded the gradient statistics in Python and published them on [GitHub](#), complete with implementation examples. These gradient statistics can be used for other GBDT

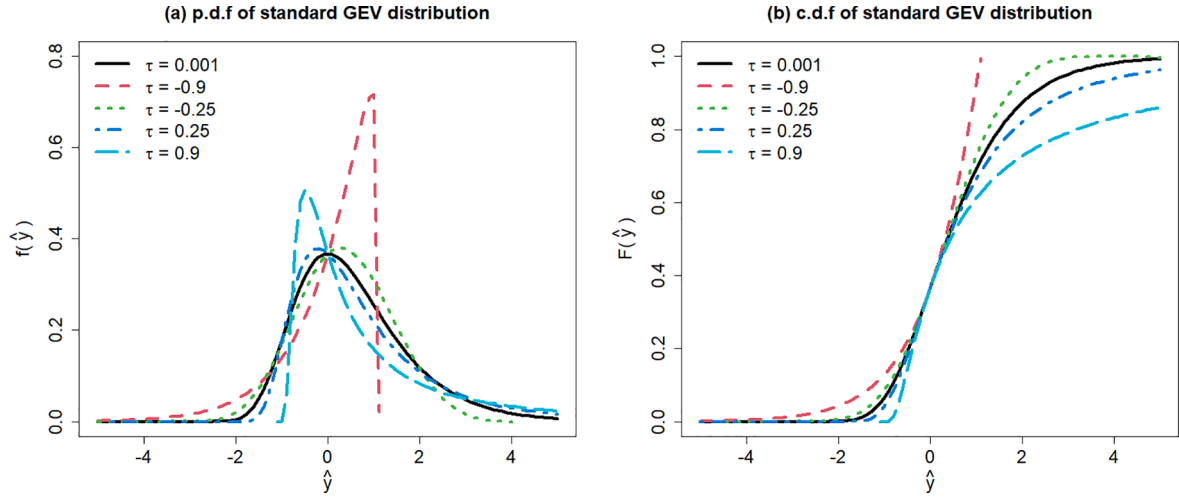


Fig. 3. a) the black line represents the type I standard GEV distribution ($\tau \rightarrow 0$), which has an unlimited domain and positive skewness. When $\tau > 0$ (type II GEV distribution), the p.d.f has negative skewness and a lower limit $-\frac{1}{\tau}$ exists on the domain. When $\tau < 0$ (type III GEV distribution), the p.d.f has positive skewness and an upper limit $-\frac{1}{\tau}$ exist on the domain. b) When $\tau < 0$ the response output of the c.d.f approaches one at a faster rate than it approaches zero and vice versa.

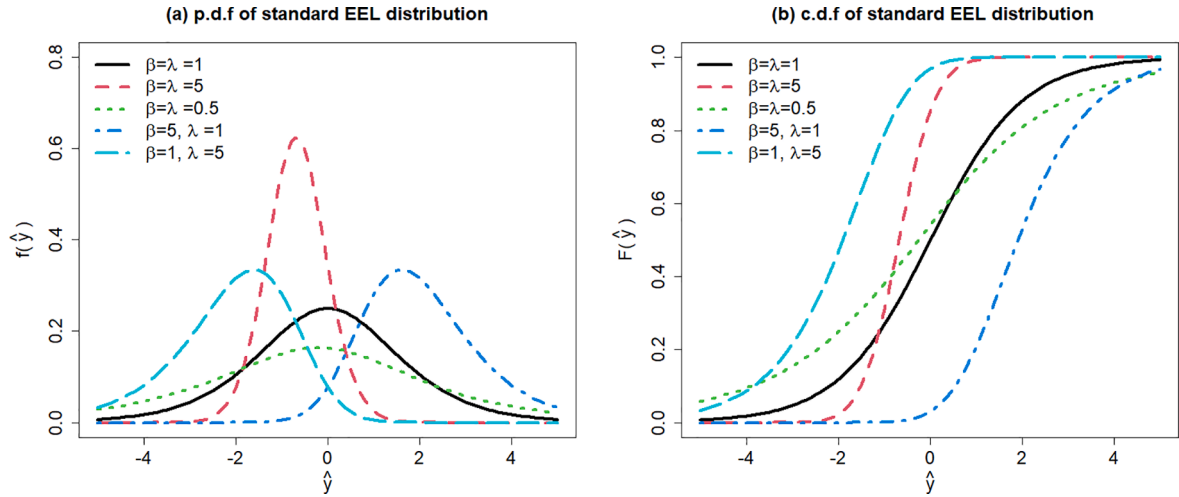


Fig. 4. a) when $\beta = \lambda$, the p.d.f. of the EEL is symmetrical with $\beta = \lambda = 1$ (black solid line) representing the standard logistic distribution. Compared to the standard logistic distribution, when $\beta = \lambda < 1$ (green dashed line) the tails are heavier and when $\beta = \lambda > 1$ (red dashed line) the tails become lighter. The standard EEL has positive skewness when $\beta > \lambda$ (dark-blue dashed line) and negative skewness when $\beta < \lambda$ (light-blue dashed line). b) When the p.d.f. is symmetric, the c.d.f. will also be symmetric, and vice versa. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

implementations, including LightGBM and CatBoost, which support custom loss functions.

5. Simulation study

The study proposes a hybrid loss that uses five hyperparameters, either a standard GEV (one hyperparameter) or EEL (two hyperparameters) distribution-based link while maintaining all baseline GBDT implementation hyperparameters. In this section, we conduct a simulation study to provide insight into the influence of hyperparameters on the predictive power of a model. In Table 1, we present seven important XGBoost hyperparameters as well as eight new ones introduced in the current study. Using sequential model-based optimization (SMBO), also known as Bayesian optimization (BO), we will optimize hyperparameters in the search spaces shown in Table 1.

5.1. Data simulation

A total of 10 artificial datasets are generated by varying the degree of

class imbalance in the Freddie Mac (FM) 60 data from our earlier study (Mushava & Murray, 2022). There is also a public version of the dataset available on Mendeley Data (Mushava, 2023). The FM60 dataset comprises 649,608 home loans, all with credit scores under 754. Among these, 18,463 defaulted within 60 months after the loan issue date. There are 78 predictor variables in the dataset, recorded at the time of loan issuance, and a binary 0/1 target variable, where 1 indicates a default within the 5-year outcome window.

A stratified random sampling method was used to create mid-sized datasets of 50 000 loans with different combinations of non-default and default cases. If we are simulating a dataset with a 1% default rate, we will randomly select 49 500 non-default and 500 default cases from the FM60 dataset. In this instance, the imbalance ratio (IR) is 99 (ratio of non-defaulting to defaulting cases).

Noting that one only needs a single feature in the data to determine how difficult it is to classify a given dataset, we calculate the highest information value (IV) in the dataset. The convention is that (Siddiqi, 2012):

Table 1
Search space for hyperparameter optimization.

Hyper-parameter	Description	Type	Search space*
XGBoost:			
max_depth	Maximum depth of a tree.	integer	[2,6]
min_child_weight	Minimum sum of case weights (or Hessians) in a tree leaf.	integer	[1,10]
subsample	The fraction of cases randomly sampled for growing a tree.	decimal	[0.5,0.8]
colsample_bytree	Fraction of predictors to randomly consider for tree building.	decimal	[0.5,1]
eta	The learning rate to shrink node weights for conservatism.	decimal	[0,1]
scale_pos_weight	Controls the balance of positive and negative class weights	integer	[1, IR]
Gamma	Minimum gain threshold for a split to be made.	decimal	[0,10]
Extras:			
Alpha (α_1)	Class weights in the cross-entropy-based loss.	decimal	[1, IR]
Delta (δ)	Class weights in the Dice-based loss.	decimal	[0,1]
Gamma1 (γ_1)	Focal parameter in the cross-entropy-based loss.	decimal	[0,5]
Gamma2 (γ_2)	Focal parameter in the Dice-based loss.	decimal	[0,5]
Pi (π)	Weight of cross-entropy and Dice-based losses in the hybrid.	decimal	[0,1]
Tau (τ)	Tail parameter in the GEV distribution-based link	decimal	[-1,0]
Beta (β)	Shape parameter in the EEL distribution-based link	decimal	[0,5]
Lambda (λ)	Shape parameter in the EEL distribution-based link	decimal	[0,5]
Loss symmetry	The symmetry of the hybrid loss function	categorical	{UF, UF1, UF0}

*A uniform distribution is used to generate random values within the search space of continuous hyperparameters.

- IV less than 0.02: no predictive variable
- IV between 0.02 and 0.1: weak variable
- IV between 0.1 and 0.3: average variable
- IV between 0.3 and 0.5: strong variable
- IV greater than 0.5: over-predictive.

Table 2 presents a summary of ten datasets that were simulated.

5.2. Data complexity

To further understand each sample's intrinsic characteristics, we apply the approach of [García, Marqués, and Sánchez \(2019\)](#). Specifically, using the k-nearest neighbour algorithm (with $k = 5$), each case in the sample is assigned to the following groupings:

- Safe: if at least 4 out of its 5 nearest neighbours belong to the same class label.
- Borderline: if 2–3 out of its 5 nearest neighbours belong to the same class label.
- Rare: if only 1 out of its 5 nearest neighbours belongs to the same class label.
- Outlier: if all the 5 nearest neighbours belong to the opposite class.

Fig. 5 summarises the complexity of the data points for the positive ($y = 1$) and negative ($y = 0$) cases. The pattern in Fig. 5 resembles that observed across various 14 real-life datasets in the experiment by V. [García et al. \(2019\)](#). That is, a high IR is associated with the positive class being outliers/rare data points. Simultaneously, when the IR is high, the negative class consists mostly of safe data points. The percentage of safe and borderline data points increases for the positive samples as IR decreases, in line with the results of the experiment by V. Garca, Marqués, and Sanchez (2019). For the negative class, however, as the sample becomes more balanced, the safe data points decrease and the number of borderline, rare and outlier data points increases. The simulated datasets are therefore representative of real-life data under different levels of class imbalance.

5.3. Hyperparameter optimisation

For each sample in Table 2, a random sample of 30 000 cases will be used for model training, 10 000 for model validation (i.e. determining optimal hyperparameters), and the remaining 10 000 for model testing (i.e. evaluating the final model). The H measure is used to evaluate model performance ([Hand & Anagnostopoulos, 2014](#); [Hand, 2009](#)). It is a coherent alternative to the widely used AUC for evaluating the discriminatory power of models.

We will utilize sequential model-based optimization (SMBO), also known as Bayesian optimization (BO) to determine the optimal combination of hyperparameters. This technique is superior to other methods such as grid search, especially when an array of hyperparameters is present ([Bergstra, Komer, Eliasmith, Yamins, & Cox, 2015](#); [Snoek, Larochelle, & Adams, 2012](#); [Xia, Liu, Li, & Liu, 2017](#)). The hyperopt module in Python will be used to implement BO using the Tree-of-Parzen-Estimators (TPE) approach, as recommended by [Bergstra et al. \(2015\)](#).

For a given candidate hyperparameter set, model training stops when the H values do not improve on the validation sample in the next 10 iterations. Once the optimal number of iterations has been determined, the hyperparameter configuration is used to score the test sample and the H measure is recorded. The BO algorithm will use the performance recorded in previous iterations to determine the next combinations of hyperparameters to consider in the next iteration.

Using's heuristics by [Owen \(2021\)](#), we set the BO algorithm to take 10 loops in the search space for each continuous hyperparameter and 15 loops for each categorical hyperparameter, for faster computation. Suppose the BO algorithm is run to search either UF, UF0, or UF1 losses,

Table 2
A summary of the simulated samples with varying IRs. There are only a few variables that have average to strong predictive power.

Dataset	#Cases	#Covariates	#Predictive variables ¹	IV ²	Default Rate	IR
Sample 1	50,000	78	10	23.32%	0.0625%	1599
Sample 2	50,000	78	13	30.10%	0.125%	799
Sample 3	50,000	78	11	24.89%	0.25%	399
Sample 4	50,000	78	11	23.00%	0.5%	199
Sample 5	50,000	78	9	38.41%	1%	99
Sample 6	50,000	78	11	35.89%	2.5%	39
Sample 7	50,000	78	11	33.18%	5%	19
Sample 8	50,000	78	10	35.06%	10%	9
Sample 9	50,000	78	10	35.14%	20%	4
Sample 10	50,000	78	10	36.94%	30.9%	2

¹ IV >= 0.02.

² Highest information value.

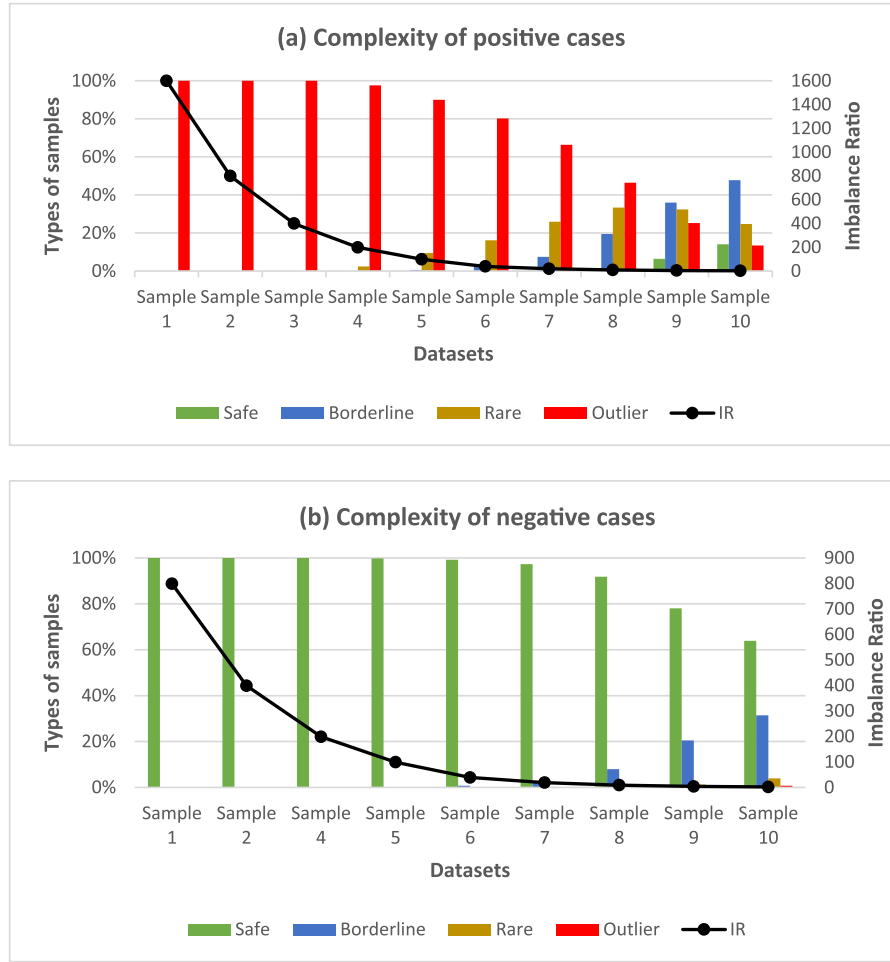


Fig. 5. Percentage of each data point type (left y-axis) and the class imbalance ratio (right y-axis) for each of the simulated samples. a) There is an association between a high-class imbalance level and default cases being outliers. When samples become more balanced, outlier default cases decrease, while rare, borderline, and safe cases increase. b) Most non-default cases are safe, but borderlines appear as the sample becomes more balanced.

each with 12 hyperparameters (7 unique to XGBoost, 5 from the hybrid loss), and using an EEL link (2 hyperparameters). In this case, the total number of iterations for the loss symmetry and the continuous hyperparameters will be $(3 \times 15) + (14 \times 10) = 185$.

Using different combinations of training, validation, and test samples in Table 2, the process is repeated five times.

5.4. Results

This simulation study aims to gain insight into the influence of hyperparameters on the predictive power of the customized XGBoost model. Therefore, for each loop of the BO algorithm, the following information was recorded: the dataset's IR, five hyperparameters from the hybrid loss, seven hyperparameters from vanilla XGBoost, a binary indicator for the type of link function ($GEV = 1$), three binary indicators for the skewness of the link (negative, positive, or symmetric), three binary indicators for the symmetry of the cross-entropy-based loss (UF, UF0, and UF1), and the H measure on the test sample.

The H measure is then set as the target variable, with the remaining variables serving as independent variables. Random forests are then applied to identify important variables (Breiman, 2001). Random forests are constructed using 1000 regression trees as base learners, and a variable's importance is determined by the average decrease in mean square error (MSE) when an independent variable is used to split a node. Fig. 6(a) presents a regression tree within the random forest, and Fig. 6(b) summarizes the importance of the variables.

The IR and learning rate are the most influential factors in the discriminatory power of the custom XGBoost. An examination of one of the component regression trees shown in Fig. 6(a) indicates an IR of 98 is the most important factor in separating the data. These results are consistent with earlier studies that have highlighted the challenges of dealing with classification tasks where a minority class represents less than 1% of the samples, as reported by Yang and Wu (2006). Accordingly, when the IR is 99 or more (e.g., Samples 5–10 in Table 2), we will refer to it as a severe class imbalance, otherwise as a non-severe class imbalance. A supplementary document is provided that analyzes how the final hyperparameters, grouped by IR severity, were selected.

6. Application

In addition to applying the hybrid losses UF, UF0, and UF1, we explore the four special cases of the loss in (11): cross-entropy (CE), CE incorporating class weights (wCE), CE with a focal parameter (fCE), and CE combined with both class-weights and focal parameter (mfCE). The 7 different losses are coupled with 4 links derived from the following standard distributions: logistic (i.e., Logit), positively skewed GEV, symmetric EEL, and positively skewed EEL. As the Logit is an instance of the symmetric EEL, we exclude pairing the hybrid loss with it. Additionally, we omit negatively skewed distributions since they are inappropriate for our classification task. Furthermore, we examine Xia et al. (2017)'s cost-sensitive link paired with a CE loss, and extend it to be compatible with the mfCE loss. In total, we explore a comprehensive set

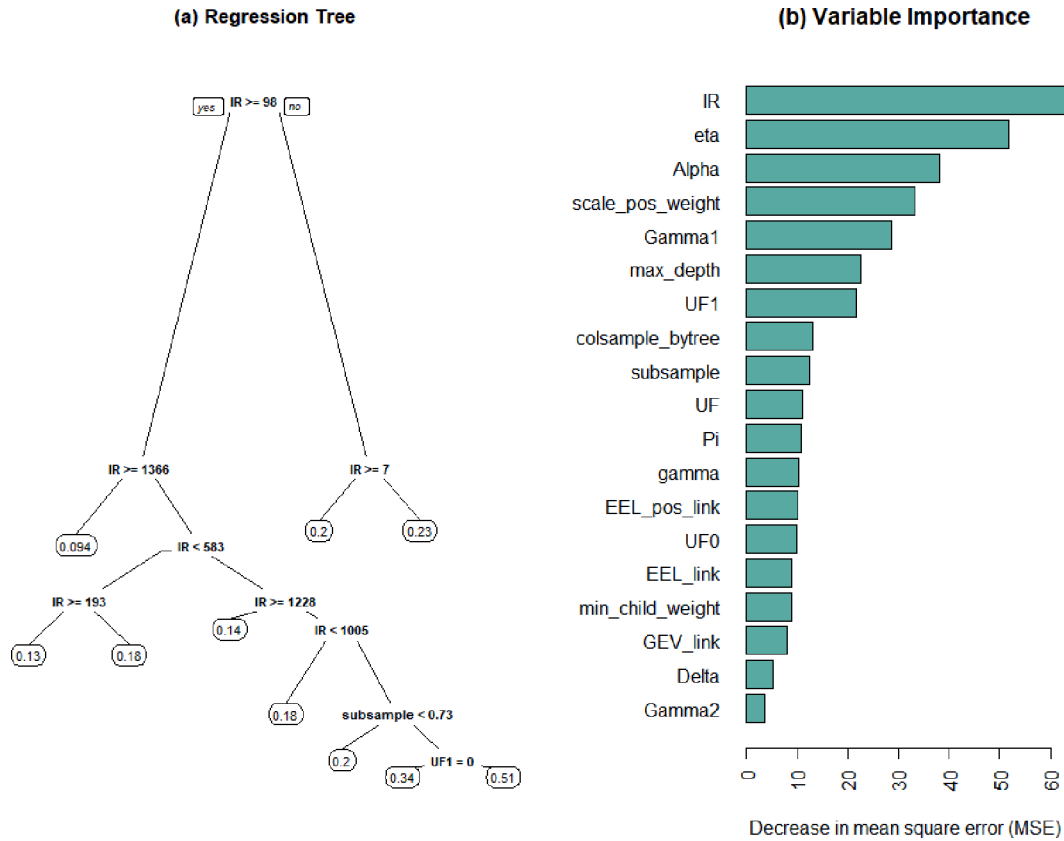


Fig. 6. a) the standalone pruned regression tree provides an interpretable picture of how *ir* and the custom *xgboost* hyperparameters influence the discriminatory power. the length of a branch is proportional to how much influence the split has on creating homogeneous groups; therefore, an *ir* of 98 is the most important factor in separating the data. b) demonstrates that among the hyperparameters in Table 1, the following influence the discriminatory power the most: *eta*, *Alpha*, *scale_pos_weight*, *Pi*, *Gamma1*, *max_depth*, and the loss symmetry.

of 27 distinct XGBoost model configurations, each built upon these various loss-link function combinations.

6.1. Datasets

We apply the models on datasets from our earlier study (Mushava & Murray, 2022): FM12, FM36, and FM60 datasets, which predict defaults in the next 12, 36 and 60 months, respectively. The datasets are also available on Mendeley Data (Mushava, 2023). As reported by Mushava and Murray (2022), at least 90% of the positive cases in the FM datasets are rare to outliers, whereas almost all of the negative cases are safe. Accordingly, default events can be viewed as rare events compared to non-default events.

The pre-processing of the data included the identification of predictive and stable covariates, their standardization, missing values imputation, and one-hot encoding. XGBoost is non-parametric, so dimension reduction techniques such as removing redundant or correlated features were not performed. The credit score variable in FM datasets is uniquely capable of predicting defaults in a given outcome window, as evidenced by its $IV \gg 0.5$ (Sousa, Gama, & Brandão, 2016). Therefore, the FM datasets used in this study correspond to loans with credit scores below a particular cut-off score to avoid having a one-dimensional model. Details of the model development, out-of-sample (OOS) and out-of-time (OOT) primary datasets used in the application are summarized in Table 3.

6.2. Benchmark classifiers

A comparison will also be made to other established classification techniques for handling class-imbalanced data in a two-class setting

Table 3

Freddie Mac primary datasets. All datasets have 78 covariates.

Data	Model development		OOS Testing		OOT Testing	
	#Cases	IR	#Cases	IR	#Cases	IR
FM12	231 065	390	232 515	389	233 791	397
FM36	159 199	163	158 784	165	606 616	113
FM60	214 064	34	230 719	34	564 408	89

(Haixiang et al., 2017). Python modules, particularly “imblearn”, “sklearn” and “keras” were used to implement these benchmarks and our code for the ensembles. A brief overview of some prominent credit scoring techniques, grouped by how they handle class-imbalanced data follows (Dastile,

- b) Cost-sensitive support vector machine (CSSVM). Cao, Zhao, and Zaiane (2013) describe the rationale of CSSVM for imbalanced data learning. IR is used as a class-weight for the minority class, and 1 for the majority class. We experimented with both linear and radial basis function (RBF) kernels. BO using the model development sample led us to favour the linear kernel due to its computational efficiency and comparable performance to the RBF kernel on our datasets.
3. Hybrid techniques:
 - a) Balanced RF (BRF) with 1000 trees (O'Brien & Ishwaran, 2019). The fraction of predictors to be used in the tree base learners is tuned.
 - b) EasyEnsemble with AdaBoost as a base learner (Liu et al., 2008). The number of samples to draw and the number of boosting iterations are tuned.
4. Heterogenous ensembles:
 - a) Stacking (Wolpert, 1992). From the techniques mentioned above for addressing class-imbalanced data in the model, we have chosen the following methods to construct a set of diverse base learners: 1b), 2a), and 3b). Additionally, we select one of the XGBoost variants as a fourth base learner. We chose an XGBoost variant whose PDs have the highest IV during model training. The BRF classifier is set as the *meta*-learner.
 - b) Bstacking is a variation of stacking proposed by Xia, Liu, Da, and Xie (2018). The idea is to create independent samples for training the base learners and *meta*-classifiers, repeating this process T times. Our stacking approach in 4a) is the special case, $T = 1$. We use the same set-up 4a) with $T = 40$ for Bstacking.

6.3. Model evaluation

Due to the large size of our datasets, the following process is repeated 30 times: we start by randomly selecting 100 000 cases from the model development dataset, and simultaneously, 50 000 cases from both the out-of-sample (OOS) and out-of-time (OOT) testing datasets. Next, we divide the model development sample into two halves using random allocation – 50% is dedicated to training, while the remaining 50% is assigned for validation purposes. After identifying the most suitable hyperparameters, we proceed to fit the models using 50 000 cases from the training subset. In the case of both the CSDNN and XGBoost variants, the validation subset is utilized for implementing early stopping. Specifically, if the H measure fails to exhibit improvement over the course of 10 iterations using the validation set of 50 000 instances, the training process is halted. The final assessment of model performance is conducted using an independent set of 50,000 cases from both the out-of-sample (OOS) and out-of-time (OOT) datasets.

This iterative model evaluation approach, incorporating diverse training, validation, OOS, and OOT samples, is specifically designed to mitigate potential biases that might arise from a single hold-out validation strategy (García, Marqués, & Sánchez, 2015). Thus, we can draw meaningful conclusions regarding the central tendency of the model performance, ensuring statistically sound and reliable evaluations. In the OOT sample, loans are issued in a future period compared to the model training sample, so it helps gauge the model's robustness to the time change.

We will also evaluate the discriminatory power of the models using the Gini index, which is a standard measure in the credit industry (Siddiqi, 2012). In binary classification tasks, discriminatory power refers to a model's ability to rank-order positive and negative cases using a numeric score, typically a probability prediction. The advantage of using the H measure and Gini index is that they are independent of a subject-based cut-off score for classifying a case. In both the Gini index and the H measure, the values range from 0 to 1.0, with higher values indicating greater discriminatory power.

In practical applications, credit-scoring models play a pivotal role by generating scores that are subsequently divided into distinct ranges,

each catering to various decision-making scenarios. Thus, our model evaluation will include assessing the accuracy and financial impact of models with high discriminatory power when they are used for decision-making based on a predetermined threshold score. Lastly, we will assess our proposed XGBoost extensions for transparency. This is an essential requirement for stakeholders with less technical backgrounds, and a crucial regulatory requirement, particularly when scoring prospective borrowers.

6.4. Results and discussion

A detailed discussion of the results is provided in the following subsections.

6.4.1. Discriminatory power

6.4.1.1. Xgboost extensions. Fig. 7 displays boxplots representing the top 10 out of 27 XGBoost variants, ranked according to their median H measure values. The values were derived from 30 model training and testing runs. To order model performance, the median was chosen because it is unaffected by outliers or skewed distributions. An outlier in performance is denoted by a filled dot, while the average performance is represented by a filled square. The horizontal dashed red lines within the boxplots represent the median of the standard XGBoost model. Additionally, to distinguish the positively skewed EEL link from its symmetric counterpart, a "+" suffix is used.

Each boxplot includes information about the class imbalance ratio (IR) in the OOS and OOT test samples during the 30 runs, as well as the p-value generated by Kruskal-Wallis's test. The assumptions of Kruskal-Wallis's test, specifically the homogeneity of variance assumption, were violated when Levene's test was applied. As a result, a significant p-value indicates that at least one model performs stochastically differently from the others, but does not necessarily indicate that the average performance differs (Kruskal & Wallis, 1952). The Mann-Whitney U test, also known as the Wilcoxon rank-sum test, is used when we are attempting to determine whether the performance of two models is stochastically different.

In the highly imbalanced FM12 OOS test dataset, among the variety of different loss functions and link pairs explored, the combination of the UF loss and the EEL + link yielded the highest median H measure (0.1639), closely followed by the CE loss coupled with the Logit link (0.1635), as well as the UF1 loss coupled with the EEL link (0.1625). When extended to OOT test datasets, it becomes evident that the CE loss together with the Logit link demonstrates the strongest discriminatory power, registering a median H value of 0.1266. It was followed by the UF loss with the EEL + link (0.1237) and the UF1 loss with the EEL link (0.1227). According to the Kruskal-Wallis test p-value, there are no statistically significant differences between the top 10 specialized loss and link function pairs for the FM12 OOS test sample. In OOT testing the FM12 datasets, Kruskal-Wallis' test p-value indicates performance differences. According to the Wilcoxon rank-sum test, with a p-value of 0.6414, there is no statistically significant difference between the vanilla XGBoost and the enhancement utilizing UF loss and the EEL + link.

In the FM36 datasets with less severe class imbalances, Kruskal-Wallis' test indicates statistically significant differences in performance among the top 10 XGBoost extensions across both the OOS and OOT datasets. Particularly noteworthy is the superiority of the UF-EEL + and UF1-EEL loss-link pairs, with OOS median H values of 16.86% and 16.76%, respectively, in contrast to vanilla XGBoost's median performance of 16.65%. This trend is similarly pronounced in OOT testing, where the UF-EEL + and UF1-EEL loss-link pairs secure median H values of 12% and 11.97%, outperforming vanilla XGBoost's fifth-ranked median H of 11.67%. Observing the FM36 OOT boxplots, it is evident that the UF-EEL + loss-link function pair achieved a substantially higher maximum H value (15.8%) compared to the vanilla XGBoost solution

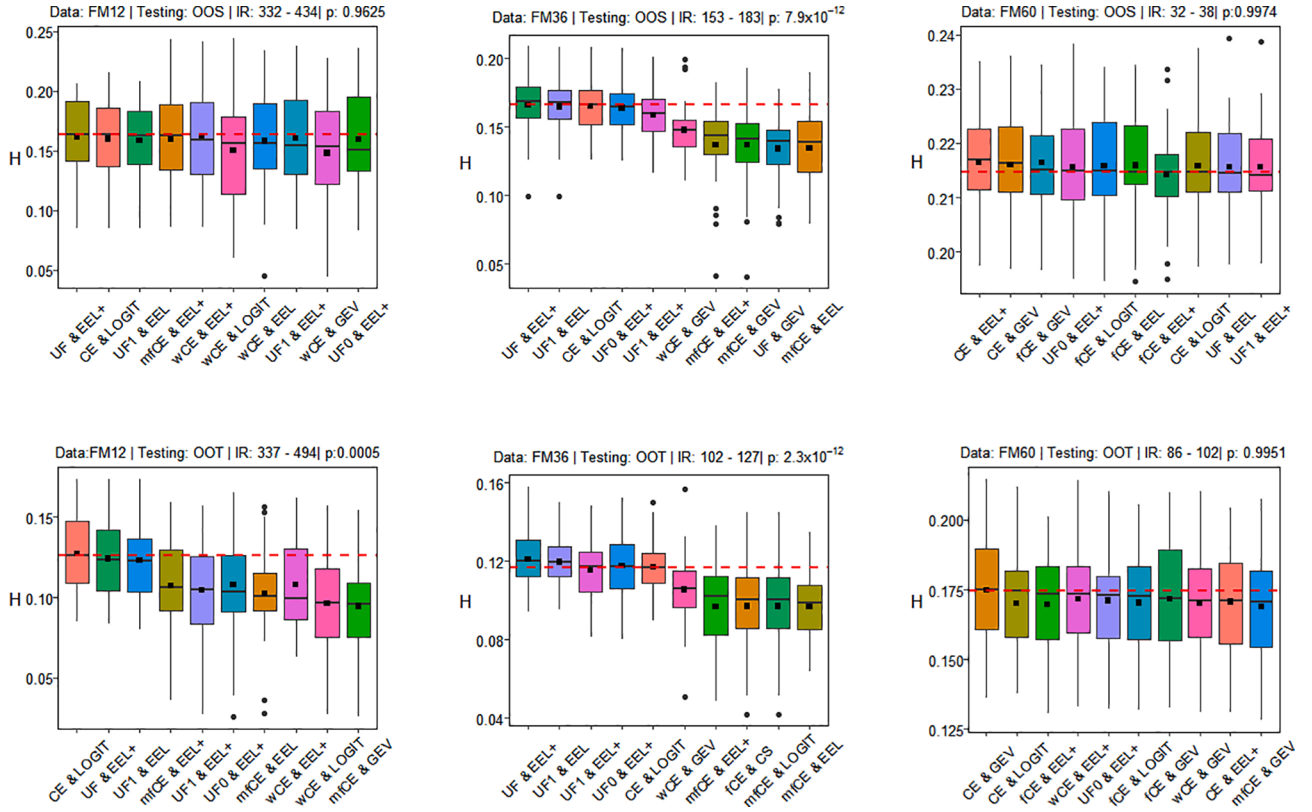


Fig. 7. Comparing the discriminatory power of the XGBoost extensions using the H measure.

(14.5%). Wilcoxon rank-sum test, p -value = 0.2618, detected no statistically significant difference between vanilla XGBoost and the tailored XGBoost using UF loss and EEL + link.

Under conditions of extreme class imbalance within the FM12 datasets, the use of the UF-EEL + and UF1-EEL loss-link pairs did not provide any noticeable advantage in terms of discriminatory power when considering median performance. However, their performance within the context of FM36 datasets has revealed promising outcomes for datasets characterized by a high imbalance ratio (IR). An intriguing finding emerges when scrutinizing the highest H values achieved on the FM12 OOS test samples. In this case, the mfCE-EEL + pair ranked fourth in terms of median performance, achieving an exceptional H-value of 24.3%, notably surpassing XGBoost's highest H-value of 21.5%. Despite XGBoost's median dominance on the FM12 OOS test samples, it shares equivalent high H-values with the UF-EEL + and UF1-EEL loss-link pairs. Fig. 7 reinforces this trend, illustrating that the custom extensions consistently achieve higher or competitive H-values in comparison to XGBoost. We posit that with meticulous refinement, the advanced loss functions introduced in this study hold the potential to outperform XGBoost's CE loss, which lacks tunability. This assertion is grounded in the fact that no further hyperparameter tuning was pursued beyond the initial Bayesian optimization stage, which employed coarse search ranges as detailed in Table 1.

In scenarios where the class imbalance is less severe ($IR < 99$), such as in the FM60 dataset, the Kruskal-Wallis test did not detect performance differences for the top 10 models, both in OOS and OOT contexts. Nevertheless, we observed that simpler loss functions, like the classic CE, were sufficient. Specifically, coupling CE with the positively skewed GEV link consistently demonstrated favourable performance in both OOS and OOT scenarios. On OOT testing, the average performance of the CE-GEV loss-link pair at 17.49%, noticeably surpassed that of vanilla XGBoost at 17.02%. Based on the observed outcomes, we find that customizing XGBoost with a GEV link offers a parsimonious alternative to the commonly used Logit link, as it introduces just a single additional

hyperparameter. Our analysis of hyperparameter optimization outcomes, as detailed in the [supplementary material](#), aligns with the findings of (Calabrese et al., 2016). In their exploration of the incorporation of a GEV link within a generalized additive model (GAM) and in the context of non-severely class-imbalanced data, Calabrese et al. (2016) determined that $\tau = -0.25$ yielded optimal empirical results. Consequently, the need for extensive hyperparameter optimization specifically for the GEV link appears to be unwarranted.

Interestingly, the cost-sensitive XGBoost (CSXGBoost) framework proposed by Xia et al. (2017) did not appear among the top 10 XGBoost variants. In our new extension, we used the CE loss with a focal parameter in CSXGBoost, represented by the fCE-CS loss-link pair, to achieve the 7th rank in OOT testing on the FM36 dataset. The proposition of Wang et al. (2020) that incorporates weighted CE (wCE) or standard focal loss (fCE) intermittently appears within the top 10 rankings but is not ranked highly. Also, the extension developed by Trisanto et al. (2021), which utilizes the modified focal loss (mfCE) alongside the Logit link in XGBoost, only obtained the 9th position on the FM36 dataset during OOT testing. The combination of mfCE loss and the GEV link, which was introduced earlier (Mushava & Murray, 2022), performed well in the special loss function case of adopting CE or wCE loss. A possible explanation for the difference between this study and the aforementioned related works is the reliance on single hold-out validation. Holdout estimators are often biased since only a portion of the data is used for training and validation (García et al., 2015). We recommend repeating model training and validation procedures multiple times, or adopting cross-validation techniques in the case of smaller datasets, in order to mitigate the bias.

In conclusion, our empirical findings present a noteworthy observation. When considering the incorporation of asymmetric loss functions to address the class imbalance, it becomes apparent that assigning the focal parameter exclusively to the rare class could potentially yield superior outcomes compared to the alternative approach proposed by Li et al. (2019) of applying the focal parameter solely to the majority class.

This distinction becomes particularly pronounced when contrasting the performance of the UF1 loss formulation, introduced in our study, with the asymmetric UF0 loss proposed by [Yeung et al. \(2022\)](#) within the realm of computer vision applications. The next section compares the discriminatory power of the top-performing XGBoost extensions to the benchmark classifiers discussed in [Section 6.2](#).

6.4.1.2. Comparison to benchmark classifiers. We compare XGBoost's discriminatory power and its top-performing customizations with the benchmarks discussed in [Section 6.2](#). XGBoost enhancements that employ UF-EEL + and UF1-EEL loss-link pairs will henceforth be referred to as XGBoost_UF_EEL + and XGBoost_UF1_EEL. These two customizations were selected for their robust performance on datasets in which there is a severe class imbalance ($IR \geq 99$). On the other hand, for datasets with non-severe class imbalance ($IR < 99$), we select the variant of XGBoost incorporating the GEV link into the CE loss, referred to as XGBoost_CE_GEV. The Gini index is also shown, which is the credit industry's standard for measuring a model's discriminatory power. We use the median as a measure of central tendency to mitigate the influence of outliers or skewed performance. In addition, the median model training time across all datasets is presented. Classifiers demonstrating the best performance are highlighted in bold text, while those demonstrating the best performance in their respective categories, including algorithm-level, data-level, hybrid, and ensemble classifiers, are underlined.

[Table 4](#) illustrates that the novel extensions introduced in this study, which could be further refined to enhance their discriminatory power, show a notable advantage over the baseline XGBoost algorithm on average. In cases of non-severe class imbalance, the XGBoost_CE_GEV variant performs significantly better than established benchmark classifiers, such as neural networks and SVMs, and sophisticated

heterogeneous ensembles. This superiority is affirmed across both the H measure and the Gini index, spanning both out-of-sample (OOS) and out-of-time (OOT) datasets. Nonetheless, XGBoost_CE_GEV required on average 50% more training time than the conventional XGBoost model.

When dealing with severe class imbalances, as observed in the FM12 and FM36 datasets, the hybrid EasyEnsemble classifier demonstrates superior performance to other classifiers. EasyEnsemble selects all instances from the minority class and combines them with the same number of cases randomly selected from the majority class. The process is repeated multiple times to generate a range of balanced training samples. Based on these balanced sample subsets, a base learner, in this case, AdaBoost, is trained, and its predictions are aggregated. In complex classification tasks with severe class imbalances, algorithmic approaches are frequently incorporated into multi-classifier systems, such as EasyEnsemble. XGBoost and its extensions implement more complex boosting algorithms than AdaBoost and demonstrate quicker training times. Consequently, the techniques proposed in this study can replace AdaBoost in the EasyEnsemble framework to improve the XGBoost variant's predictive capacity in severe class imbalances, where algorithm-level adjustments were less effective than hybrid classifiers.

The performance of the model was the worst when data pre-processing techniques were applied. In addition, heterogeneous ensemble classifiers such as Bsteking require significant computational resources. Also of note is that the Gini index and H measure consistently rank models according to their discriminatory power. An empirical study by [Lessmann, Baesens, Seow, and Thomas \(2015\)](#) found a 93% correlation between the area under the receiver operating characteristic curve (AUC) and the H measure. The AUC is proportional to the Gini index: $AUC = 1/2(Gini + 1)$. The next section examines the relationship between discriminatory power, and classification accuracy, which is related to profitability.

Table 4

Benchmarking XGBoost and its top-performing extensions to selected data-level, hybrid methods, and ensemble methods for handling class imbalanced data. Values shown are median (in per cent) performance across 30 validation runs.

Metric	Classifier	FM12		FM36		FM60		Runtime (sec)
		OOS	OOT	OOS	OOT	OOS	OOT	
H	Algorithm-level:							
	XGBoost	16.35	<u>12.66</u>	16.65	11.67	21.46	17.46	<u>4.40</u>
	XGBoost_CE_GEV	0.00	0.00	9.54	7.68	<u>21.62</u>	<u>17.51</u>	6.99
	XGBoost_UF_EEL+	<u>16.39</u>	12.37	<u>16.86</u>	<u>12.00</u>	21.37	16.78	10.10
	XGBoost_UF1_EEL	16.25	12.27	16.76	11.97	21.25	16.79	11.25
	CSDNN	13.40	5.04	14.60	11.30	20.70	17.10	39.10
	CSSVM	13.40	5.04	14.60	11.30	20.90	17.30	37.80
	Data-level:							
	SMOTE_RF	<u>6.50</u>	3.56	<u>6.89</u>	<u>5.94</u>	11.62	<u>7.98</u>	<u>105</u>
	SMOTE_ENN_RF	6.04	<u>4.30</u>	6.80	2.75	<u>12.70</u>	4.24	117
	Hybrid:							
	BRF	19.57	11.63	18.18	11.95	<u>19.74</u>	<u>10.85</u>	15.5
	EasyEnsemble	<u>21.8</u>	<u>13.94</u>	<u>20.05</u>	<u>14.50</u>	18.88	10.39	<u>4.6</u>
	Heterogeneous ensembles:							
	Stacking	10.88	8.31	14.14	9.89	14.92	7.78	444
	Bsteking*	<u>19.99</u>	<u>9.78</u>	<u>17.66</u>	<u>12.13</u>	<u>18.09</u>	<u>13.02</u>	18 998
Gini	Algorithm-level:							
	XGBoost	<u>45.87</u>	<u>40.09</u>	46.57	40.37	54.20	48.00	<u>4.40</u>
	XGBoost_CE_GEV	45.81	39.85	<u>47.04</u>	40.46	54.25	47.35	6.99
	XGBoost_UF_EEL+	0.00	0.00	27.65	25.63	<u>54.44</u>	<u>48.04</u>	10.10
	XGBoost_UF1_EEL	45.49	39.59	46.98	<u>40.67</u>	54.05	47.60	11.25
	CSDNN	36.80	18.20	42.50	39.10	52.40	47.40	39.10
	CSSVM	36.80	18.20	42.50	39.10	53.10	47.70	37.80
	Data-level:							
	SMOTE_RF	22.1	17.0	28.0	<u>27.1</u>	39.7	<u>33.2</u>	<u>105</u>
	SMOTE_ENN_RF	<u>23.1</u>	<u>21.1</u>	<u>30.1</u>	16.8	<u>43.0</u>	23.1	117
	Hybrid:							
	BRF	49.7	37.5	50.0	41.7	<u>52.2</u>	<u>40.0</u>	15.5
	EasyEnsemble	<u>50.7</u>	<u>40.7</u>	<u>51.4</u>	<u>44.6</u>	51.3	35.2	<u>4.6</u>
	Heterogeneous ensembles:							
	Stacking	35.3	31.8	41.8	38.1	45.5	33.6	444
	Bsteking	<u>47.2</u>	<u>34.3</u>	<u>48.7</u>	<u>41.9</u>	50.3	42.6	18 998

*Model training and testing repeated only 5 times due to intensive computational time.

6.4.2. Classification accuracy and profitability

To gauge the potential business value of employing scorecards with higher discriminatory power, we examine classification accuracy and related misclassification costs. For each FM dataset, we select XGBoost and its most effective variant, as well as the most effective algorithm-based, data-based, and hybrid benchmarks. We pursue the same model evaluation strategy outlined in Section 6.3, but we use the 50 000 cases from the OOS test data to determine a cut-off score c that maintains the training sample default rate. A case in the OOT test data, which more closely resembles real-life data, is classified as positive ($y = 1$) if the model-predicted probability is greater than c , otherwise classified as negative ($y = 0$).

From the confusion matrix, we calculate the following metrics: false positive rate (FPR), false negative rate (FNR), Matthew's correlation coefficient (MCC), and F1 score. The FPR is the percentage of positive cases misclassified as negative by the model. In contrast, the FNR is the fraction of positive cases that are incorrectly classified as negative by the model. The F1 score, which is equivalent to the Dice score in equation (16), ranges from 0 to 1, with 1 indicating the accurate classification of all positive cases. F1 scores, however, ignore the correct classification of negative cases. The MCC, whose values range from -1 to 1 , provides an unbiased measure of classification accuracy of both classes when the datasets have a significant class imbalance (Chicco & Jurman, 2020). An MCC of zero represents the expected value of a coin toss, whereas -1 and 1 indicate perfect misclassification and classification, respectively.

To assess whether more accurate models are profitable, we examine misclassification costs, as done by Lessmann et al. (2015). Credit scoring is characterized by the low opportunity cost of incorrectly classifying negative cases as positives, that is, the FPR error, as opposed to the cost or loss of incorrectly classifying a positive case as negative. In the first 12 months, which is the target of the FM12 dataset, we assume the ratio will be 1:5, and on a longer scale, such as FM36 and FM60, we assume the ratio will be 1:10. The results of the analysis are presented in Table 5.

Table 5 shows the MCC does not apply to the benchmark classifiers in the FM12 datasets. This is because the models classify all positive cases as negative. Consequently, a zero F1 score is reported. Despite EasyEnsemble's robust discriminatory power, it could not detect positive cases in OOT samples. Poor calibration may be responsible for this problem, namely, predictions are not accurately aligned with observed outcomes. There is no correlation between calibration and discriminatory power, and further post-processing is often required to align predictions with actual outcomes (Bequé, Coussement, Gayler, & Lessmann,

2017). The results indicate that there is no single model that is suitable for all performance metrics and performs optimally across them.

Across all datasets, XGBoost and its variants exhibit the highest MCC and F1 values. The FPRs, however, are lower due to poor predictions for the negative cases compared to the benchmarks. In contrast, XGBoost and its variants have relatively low FNRs due to their higher classification accuracy on positive cases. Low FNRs have significant implications for financial applications such as credit scoring since misclassifying positive cases is a serious error. As an example, denying credit to a potentially good client, although regrettable, is more manageable given the larger number of potential borrowers. In contrast, extending credit to a high-risk customer results in considerable financial losses. Benchmark classifiers are therefore susceptible to low FPR errors but incur substantial misclassification costs due to high FNR errors.

To determine whether the FNRs of the top two classifiers are statistically significant, we conducted a Wilcoxon rank-sum test. There were no statistically significant differences in the FNRs of XGBoost and XGBoost_UF_EEL+ in cases where the IR exceeded 99, namely FM12 (p -value = 0.4765) and FM36 (p -value = 0.2559). A significant statistical difference was observed between the FNRs of XGBoost and XGBoost_CE_GEV in the FM60 datasets, with a p -value of 0.0166. Furthermore, XGBoost_CE_GEV has a superior discriminatory power when the IR is lower than 99, implying that its implementation is likely to result in a tangible financial benefit.

6.4.3. Model transparency

The use of machine-learning models has gained increasing interest among banks in recent years (Dastile et al., 2020). However, advanced machine learning techniques present a significant challenge because of their inherent "black box" nature, which makes it difficult to understand how predictions are generated. For banks, credit providers, and other institutions that prioritize accurate predictions, while seeking insight into their inner workings, this concern is particularly pressing (Haixiang et al., 2017). Moreover, many countries impose regulatory stipulations demanding interpretability in credit scoring models, particularly for assessing new credit applications.

GBDT techniques such as XGBoost rely on intuitive decision trees as base models, which can readily be extracted and interpreted. Based on our hyperparameter optimization, we determined that shallow decision tree base learners are optimal, which makes it easier to interpret the trees. Fig. 8 illustrates a representative base tree learner from XGBoost_CE_GEV, demonstrating its interpretability. Based on this base tree, a client with a credit score below 678.5 and more than one obligating borrower for the mortgage (indicated by Single borrower = 0) receives a score of -0.0058 . To determine the probability of default, the component decision tree base model scores are summed up and fed into

Table 5

Median performance of selected models in terms of classification accuracy across 30 runs. Except for the misclassification costs, the values shown are percentages.

Dataset/ Classifier	Performance metric				
	MCC	F1	FPR	FNR	Cost*
FM12					
XGBoost	0.90	1.10	0.97	97.83	4.9012
XGBoost_UF_EEL+	0.74	0.97	0.44	98.49	4.9288
CSDNN	n/a	0.00	0.20	100.00	5.0020
EasyEnsemble	n/a	0.00	0.07	100.00	5.0007
SMOTE_ENN_RF	n/a	0.00	0.11	100.00	5.0011
FM36					
XGBoost	10.1	5.12	1.97	90.94	9.1150
XGBoost_UF_EEL+	8.48	5.20	2.22	90.77	9.0984
CSDNN	1.13	1.10	0.17	99.33	9.9347
EasyEnsemble	2.77	2.94	0.30	99.98	9.9984
SMOTE_ENN_RF	-0.05	0.04	0.03	99.98	9.9984
FM60					
XGBoost	6.62	5.99	7.67	75.85	7.6620
XGBoost_CE_GEV	6.04	5.27	10.73	70.95	7.2023
CSSVM	2.52	3.66	2.00	94.57	9.4768
BRF	2.52	3.66	2.00	94.57	9.4768
SMOTE_RF	1.10	2.22	1.23	97.61	9.7737

*Assuming it costs 5 times to misclassify positive cases in the first year, and 10 times afterwards.

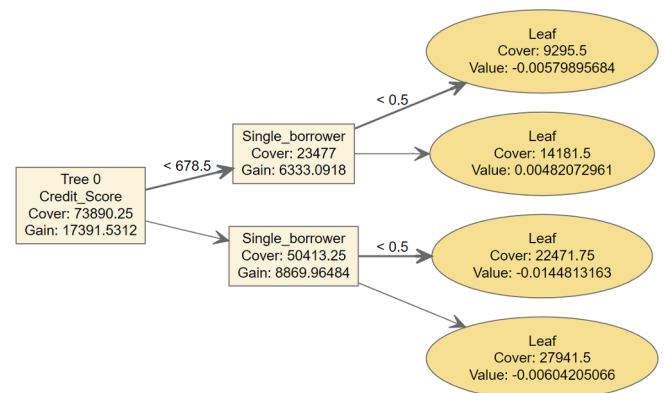


Fig. 8. An extract of the first decision tree-based learner based on the XGBOOST_CE_GEV variant from the FM60 dataset. The cover is the sum of Hessian, representing how many cases fall into that node. Gain is the information gain (8) for the split, representing the importance of the variable.

the link function.

As the number of decision trees in baseline models increases, analyzing each component tree becomes increasingly challenging. To address this issue, the baseline implementation of XGBoost incorporates contemporary techniques such as SHAP (Shapley Additive Explanations). The SHAP method can be used to assess the contributions of each feature to predictions (Lundberg & Lee, 2017). Fig. 9 shows the SHAP values of the six most influential predictors in the XGBoost_CE_GEV variant when applied to FM60 training data. Positive SHAP values indicate a higher likelihood of default, while negative values indicate the opposite. A credit score, for example, has a nonlinear relationship with default, where in general, the higher a credit score, the lower the default rate. Blue markings indicate that other factors are influencing the predictions at that point. Based on the dummy 0/1 variable for a single borrower, it is evident that a higher default rate is associated with loans with only one borrower liable to repay the mortgage, as opposed to loans with more guarantors. Other variables in Fig. 9 also exhibit logical trends and can be interpreted similarly. Our prior study provides a comprehensive description of the predictor variables (Mushava & Murray, 2022).

7. Conclusions, limitations and future research

7.1. Conclusions

This paper presents a novel flexible loss function for binary classification based on the popular XGBoost implementation of gradient-boosted decision trees (GBDT). This loss function can specify Dice-based and cross-entropy-based loss functions, and their hybrids, and

can be symmetric or asymmetric. For the link functions in the losses, there are two options: either one drawn from the generalized extreme value (GEV) distribution or an exponential logistic (EEL) distribution. We examined 27 competing XGBoost variants and applied them to a publicly available Freddie Mac mortgage loan database.

The novel extensions introduced in this study demonstrate a notable advantage over the baseline XGBoost algorithm. When the class-imbalance ratio (IR) is not severe, usually less than 99, cross-entropy loss paired with a link derived from a standard GEV distribution performed significantly better than established benchmark classifiers, such as neural networks, support vector machines, and sophisticated heterogeneous classifiers. In the case of severe class imbalances, typically IRs between 99 and 200, the more advanced loss function that combines the symmetric loss function with a link derived from a positively skewed standard EEL distribution performs materially better than the vanilla XGBoost. Across both out-of-sample and out-of-time datasets, the proposed extensions demonstrated superiority both for the H measure and the Gini index. In extreme class imbalance ratios, usually greater than 300, we did not find a clear advantage to using a more advanced loss function. Because of the hybrid loss function's flexibility, we expect it to perform better at extreme levels of class imbalance with fine-tuning.

Finally, we have found that the proposed extensions decrease the misclassification costs associated with incorrectly classifying positive cases as negative when the class imbalance is not severe, which has a financial benefit for credit scoring. Furthermore, GBDTs such as XGBoost are relatively transparent in comparison to other machine learning techniques, which is desirable for financial applications. Our working code and model output are publicly available on our [Github](#) repository to facilitate reproducibility and extensibility.

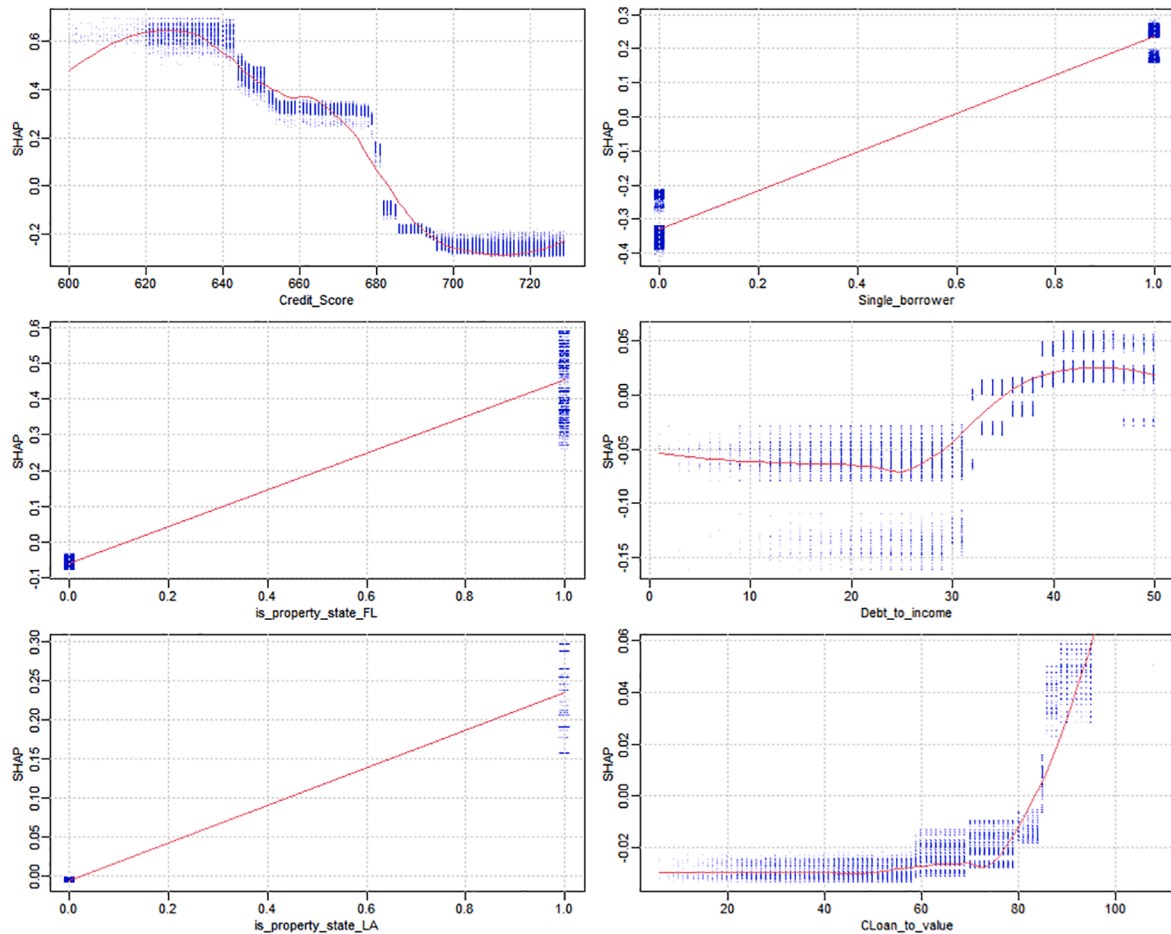


Fig. 9. The contribution of the top 6 predictors towards the target prediction in the XGBOOST_CE_GEV variant in the FM60 training dataset.

7.2. Limitations

Our study has limitations, primarily due to data resampling and hyperparameter optimization. To expedite computational processes and mitigate random effects, we selected 50,000 cases each for model development, validation, out-of-sample testing, and out-of-time testing, and repeated the process 30 times. In the case of hyperparameter optimisation, we repeated the process five times over course search ranges. While increasing efficiency, this pragmatic approach raises questions regarding the representativeness of the samples and hyperparameters.

Sample sizes and repetitions can be increased for more robust results. Another option would be to examine other data resampling techniques such as cross-validation and bootstrapping. When the class-imbalance ratio was extreme, we did not find any noticeable benefit from the advanced loss function. Considering that our proposed advanced loss functions include other established loss functions as special cases, fine-tuning the hyperparameter search range should assist in achieving at least the same performance as the base models.

Our method of central tendency assessment relies on the median, and we use out-of-time testing. Thus, any outliers, skewness, or over-optimistic results should not significantly influence our conclusions. Furthermore, to ascertain the significance of performance disparities, we incorporated consistent statistical tests into our analysis. As a result of this study's model evaluation approach, the validity of our findings is enhanced.

7.3. Future research

Our future research endeavours will extend the scope of this study by exploring a variety of avenues, including:

- Enhancing class-imbalanced data modelling: we intend to integrate GBDT models with advanced loss functions within hybrid classifiers, such as EasyEnsemble, as well as heterogeneous ensemble classifiers.
- Broadening the application of advanced loss functions: we also intend to investigate the applicability of advanced loss functions in diverse machine learning methodologies such as artificial neural networks, contemporary data augmentation techniques such as GANs, as well as other GBDT implementations such as LightGBM.
- Targeting diverse performance metrics: instead of focusing solely on optimizing discrimination power, our future research will explore alternative performance metrics such as accuracy and/or profitability.
- Evaluating effectiveness in different scenarios: not limited to credit scoring, we also intend to apply the advanced loss functions to complex datasets with small disjuncts, overlapping classes, mixed types of predictors, and multi-class targets.

We hope that by focusing our future research in these directions, we will be able to theoretically and practically improve the use of advanced loss functions in machine learning.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The manuscript provides information on how to access [supporting data](#) and code.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eswa.2023.121876>.

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